CS5012

DEPARTMENT COMPUTER SCIENCE AND ENG., I I T PALAKKAD

CS5012: AI for Cybersecurity

Mid-Semester Examination, 5th March 2026

Time: 02.00 PM – 04.30 pm **MARKS:30**

Instructions:

(i) Answer all questions.

(ii) Plagiarism/Cheating will lead to Fail Grade

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1. Differentiate between Trojan, Rootkit and botnets. 3 Marks

2. Differentiate between static malware analysis and dynamic malware analysis. 3 Marks

3. What could be the possible motivations of cyber criminals in carrying out cyber-attacks. 4 Marks

4. Explain in detail the steps to perform static malware analysis on windows PE files. 5 Marks

5. Design your own malware detection framework for either PE/ELF datasets shared with your during class assignments. This time you need to use a model different from what you have submitted in the assignment. Report accuracy, inference time (one check), F-1 and number of False positives. Provide the rationale behind the choosing this particular model and number of data samples used for training/testing. 15 Marks

END OF PAPER